

AIoT InnoWorks 2026 · Semi-final

Edge-AI Thermal Safety Monitor for EV Battery Packs

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We look for the failing cell before it becomes a fire.

Photo: lithium-ion pack destroyed by thermal runaway (JAL 787 APU battery) — NTSB, public domain



The problem

A pack does not fail all at once.

One weak cell heats up first.

Once thermal runaway starts, it feeds itself.

~31%

of EV fires start while the vehicle is parked

~25%

happen in underground car parks

1.5–12M

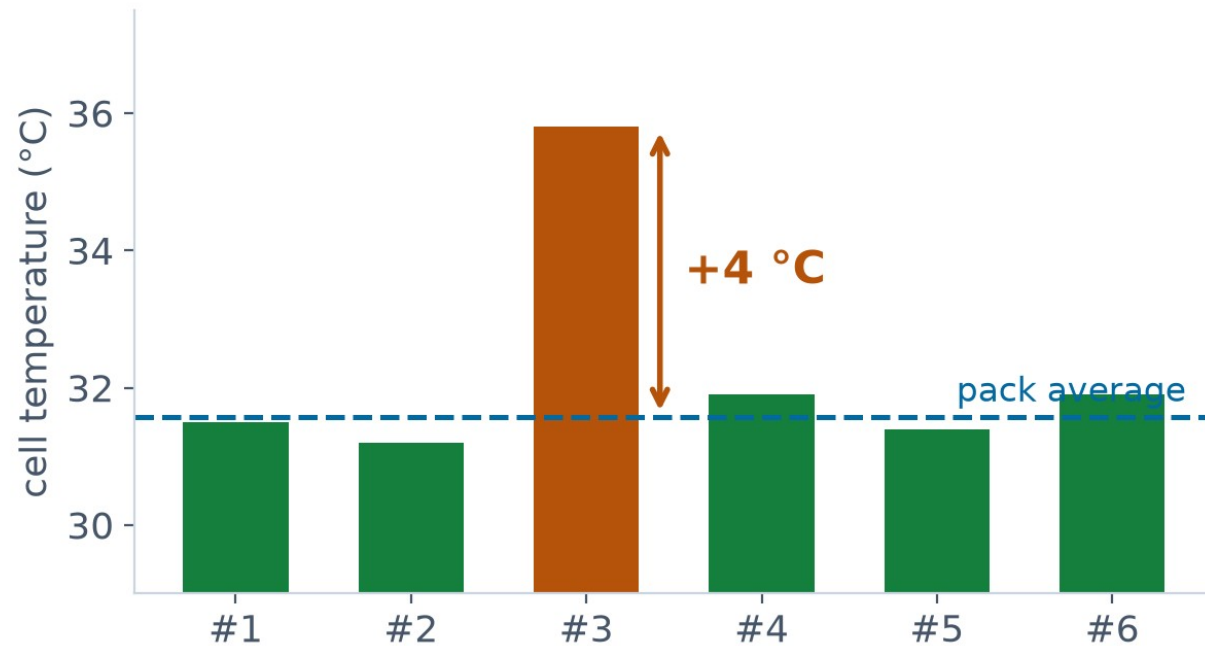
VND to replace one e-motorbike pack

Photo: vehicle fire in a car park (not an EV) — Jim Heaphy, CC BY-SA 3.0 · Fire statistics: EV FireSafe



Every BMS already has a 60 °C cut-off. So why AI?

Because the dangerous signal appears long before any absolute limit.



FIXED THRESHOLD

Verdict: all normal

Nothing is near 60 °C.

OUR EDGE AI

Verdict: abnormal cell

Cell #3 is +4 °C above the rest, under the same load.

The model reads relations between cells — not absolute degrees.

Our solution

A low-cost module that watches every cell and thinks on the device.

1 DETECT

An autoencoder on the ESP32-S3 flags the odd cell every second. No cloud needed.

2 PREDICT

A cloud model reads each charge curve and reports battery health and remaining life.

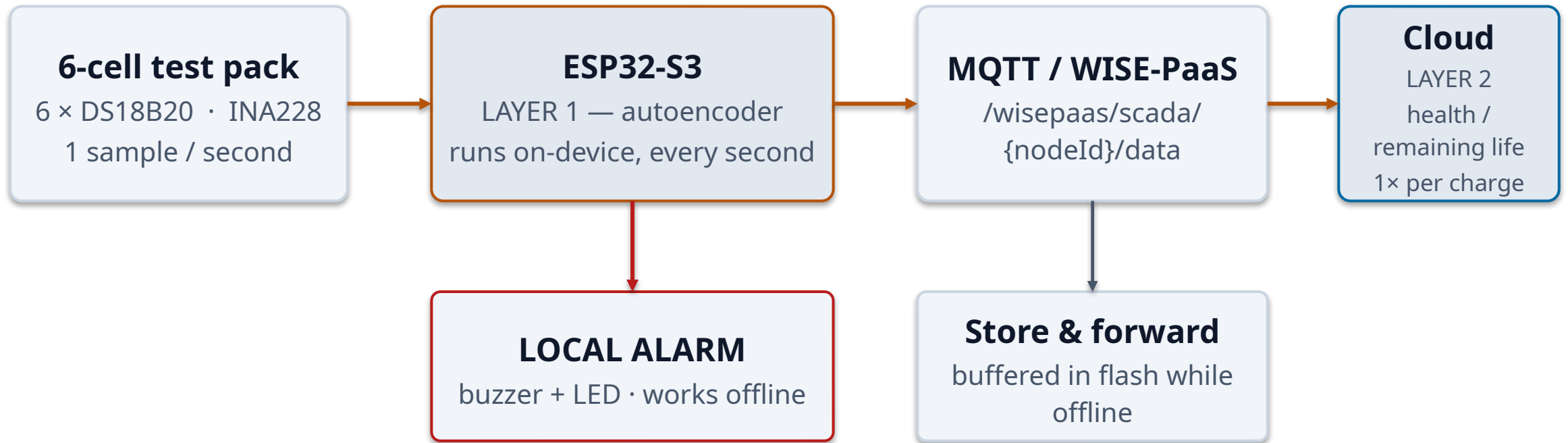
3 ACT

A 4-level local alarm: buzzer and LED fire even with the network down.



System architecture

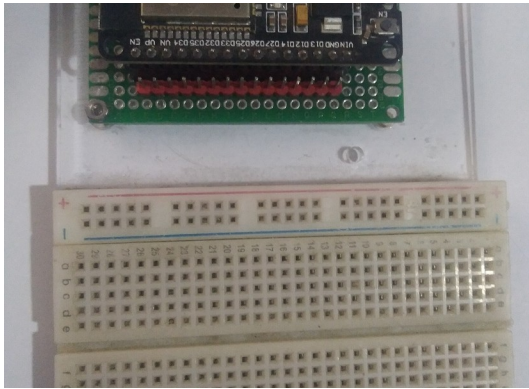
Two AI layers, split by how fast each answer is needed.



Safety never depends on the network. The device detects and alarms on its own; the cloud adds dashboards and long-term prediction.

The hardware we built

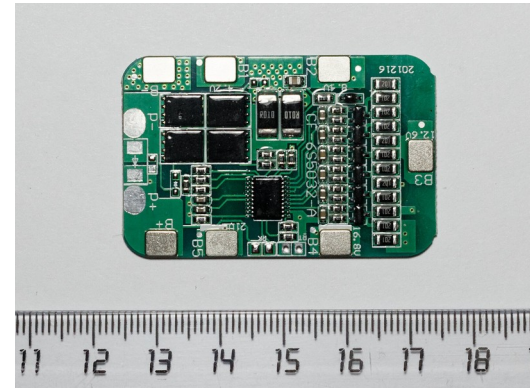
A 6-cell 18650 test pack — every cell has its own probe.



ESP32-S3-DevKitC-1
vector instructions for
ML



6 × DS18B20
one probe per cell



6S BMS with balancing
the part that cuts out



6 matched 18650 cells
same batch, same age

Faults are induced safely: a ceramic power resistor warms one cell. Firmware hard-cuts the heater at 60 °C. No cell is ever abused.

Layer 1 — the odd-cell detector

Runs on the microcontroller itself, every second, with no internet.

It reads relations, not degrees

- how far it sits from the pack average
- whether it heats up faster than the others
- slow drift (ageing) vs. sudden jump (fault)

It alarms only if the signal persists for half a minute — noise alone never triggers it.

tiny

about 1 KB
of flash

1 Hz

inference rate
on the device

offline

detection needs
no network

60 °C

hard threshold,
outside the AI

If the model is ever wrong, the hard threshold still fires. That line is never removed.

What we have tested so far

A lab pack of similar cells, plus public datasets from real packs — not an EV pack in service.

Sensors and data path, on the bench	live on the dashboard, buffered while offline
Alarm logic, full bench test	all four levels correct, even with the AI disabled
Public dataset, injected cell fault	detected reliably
Independent real pack, unseen in training	faulty runs separate clearly from healthy

These are results on the bench and on public datasets. Warming a cell on our own pack, and measuring an EV pack in service, are the next two steps — we have not done them yet.

The test we failed — and what it taught us

We report this because it is the strongest engineering result we have.

1 · FAILED

On a real pack we had never trained on, the model alarmed on a healthy run.

Worse than on the faulty one.

2 · DIAGNOSED

The absolute inputs — ambient, current, charge level — had drifted far outside the training range.

They are the same for every cell, so they lifted all cells at once.

3 · FIXED

Dropped them. Kept only cell-to-cell relations.

Fewer false alarms, better detection, and the model now transfers to unseen packs.

This is also what makes mass production possible: one identical model ships in every unit.

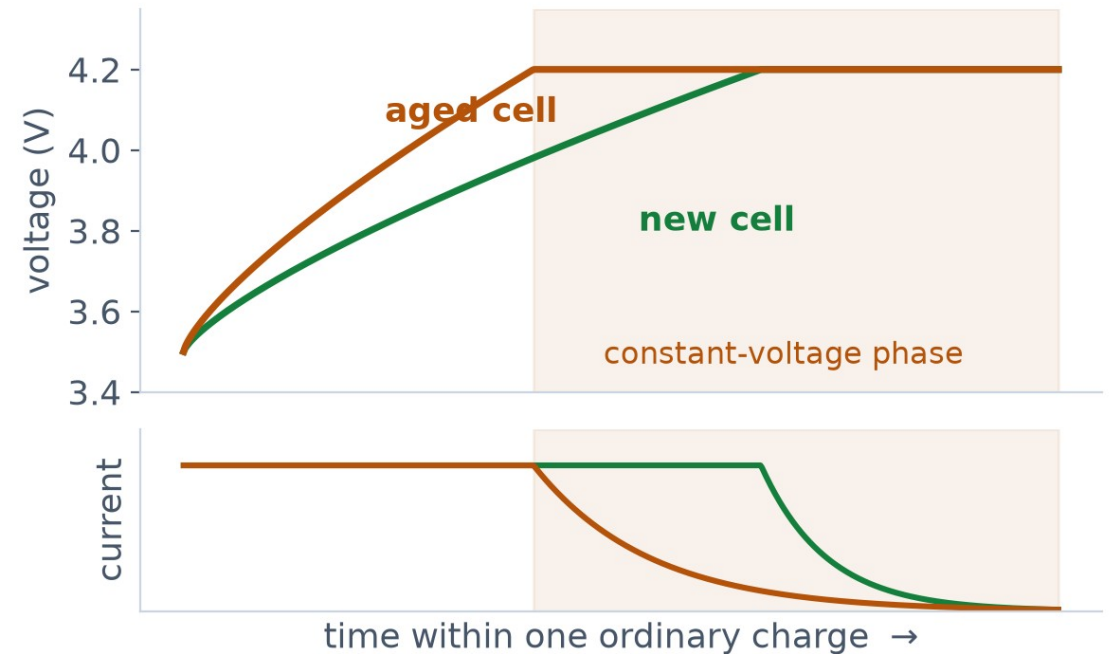
Layer 2 — how long will this pack last?

You cannot deep-discharge a pack every week. But everyone charges every day.

Read the shape of the charge curve.

An aged cell reaches the voltage limit early, then stays in constant-voltage far longer.

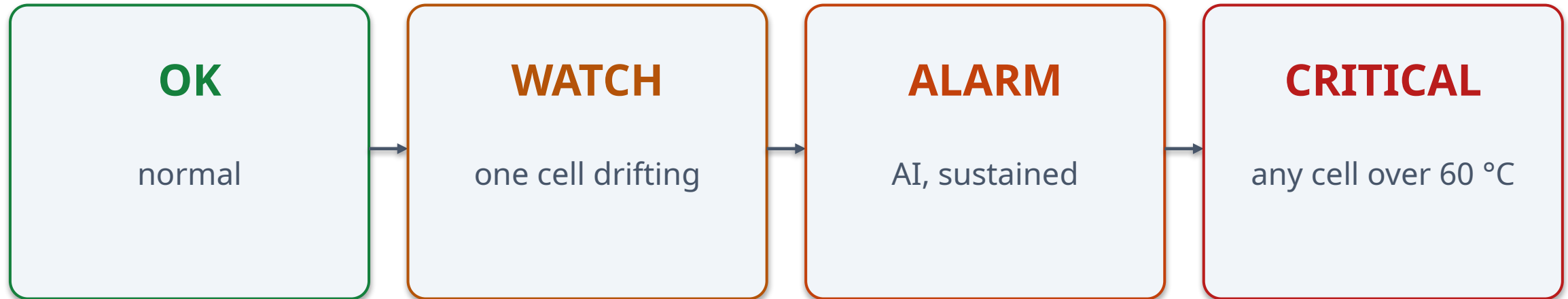
Measured during an ordinary charge — nobody has to do anything.



Layer 2 runs in the cloud on purpose: improving the model must not require reflashing every vehicle.

Four alarm levels, not one buzzer

"Keep an eye on it" and "cut the charger now" are different orders.

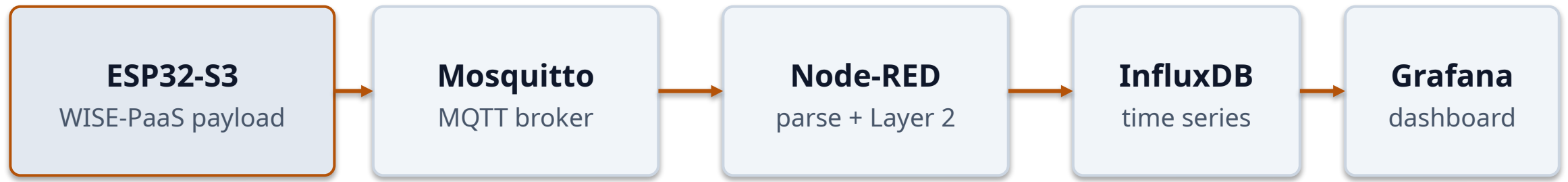


SENSOR LOST is blue, not red.
"I cannot see the pack" ≠ "the pack is dangerous".

Mute exists — and mute never kills the light.
A buzzer you cannot silence gets its wires cut.

Where the system runs today

We already speak the WISE-PaaS protocol — we are simply hosting the layers ourselves.



topic `/wisepaas/scada/{nodeId}/data` — the same format as Advantech's own Edge SDK



Our own Grafana dashboard — data shown is simulated for this screenshot

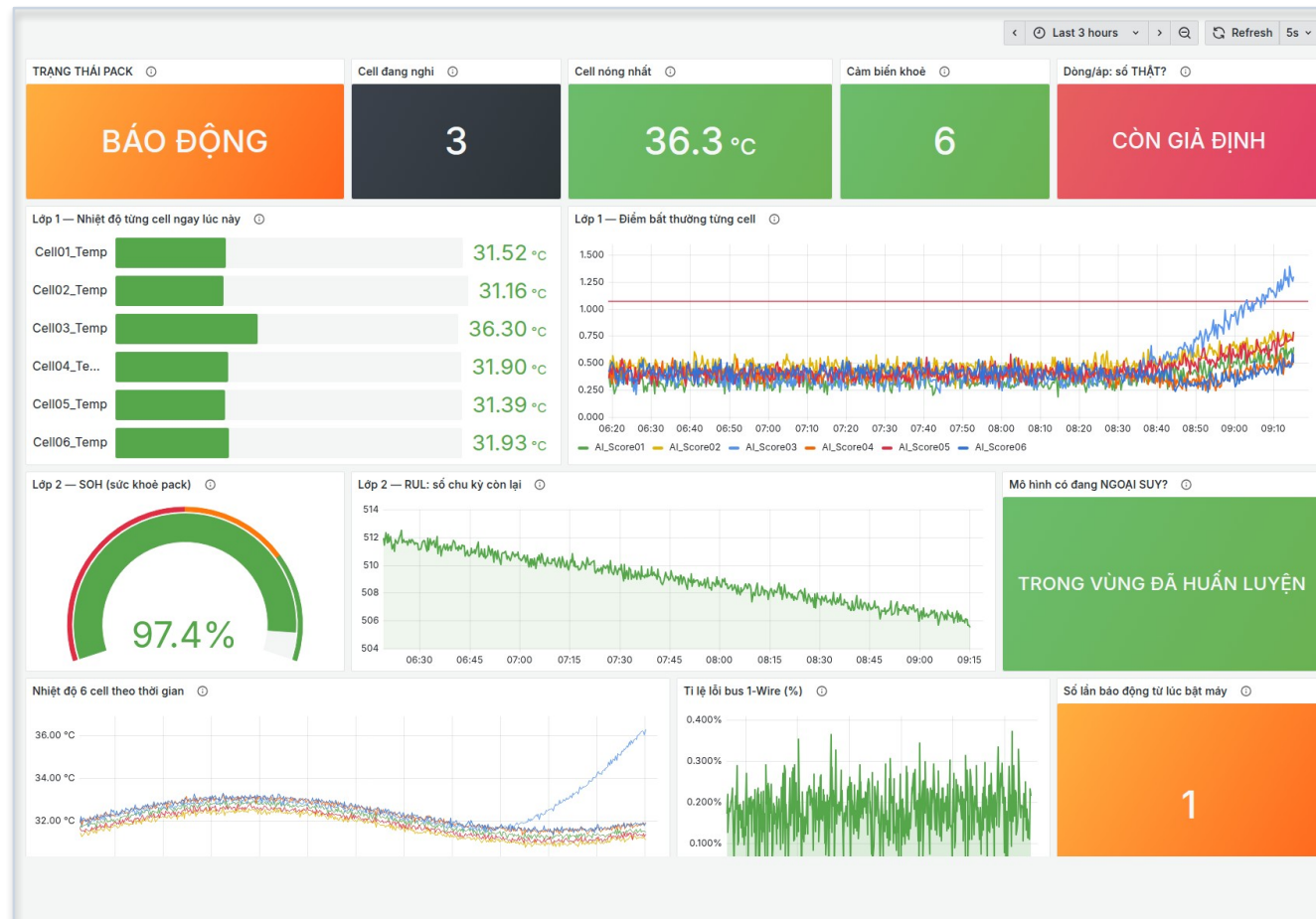
Why this stack?

These are the same components WISE-PaaS is built on.

Nothing here is a workaround we would have to throw away.

What the operator actually sees

Three seconds to answer: is the pack safe, which cell, and can I trust the number?



Top row first

Pack state, suspect cell, hottest cell, sensor health — no chart reading required.

Cell #3 stands out

The bar gauge is a spatial comparison, so the odd cell is obvious at a glance.

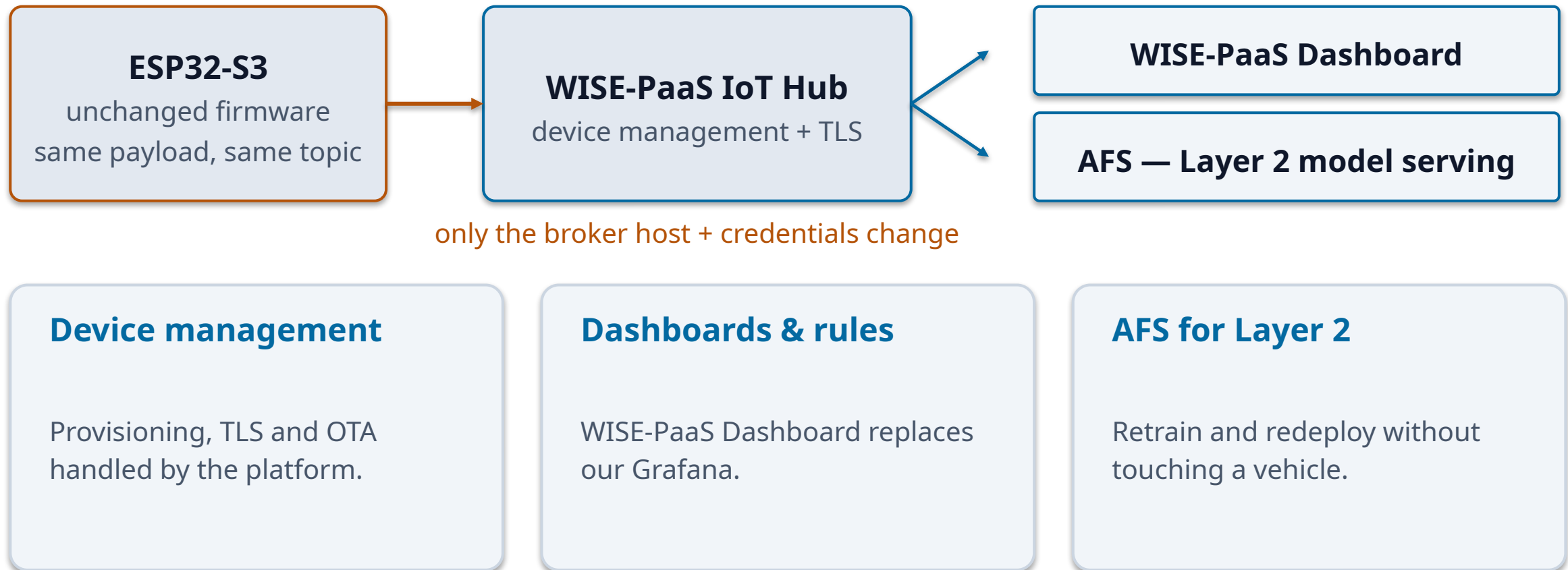
Honest by design

Numbers that are still assumptions label themselves on screen.

Our own Grafana dashboard. The data in this screenshot is simulated to show the scenario.

With a WISE-PaaS account

A config change, not a rewrite — the firmware does not change by a single line.



Live demo — 90 seconds

- 1 The dashboard is streaming live cell temperatures.
- 2 We unplug the network in front of you.
- 3 We warm one cell with the resistor.
- 4 **The module detects it and sounds the alarm — still offline.**
- 5 Network back: the offline data lands on the dashboard, at the right timestamps.

Edge AI, local alarm and store-and-forward — all proven in one run.

Photo: e-scooter charging station — Wikimedia Commons, CC BY-SA 3.0



Who pays for this

Fleet operators and battery-swap networks — where one company owns the packs.

Owner = payer

The operator owns the packs, so whoever loses money on a dead pack is whoever can buy the device.

Planned, not urgent

Pack replacement becomes scheduled maintenance instead of an emergency.

No SIM needed

Packs come back to a station daily — station Wi-Fi is the uplink.

Individual riders are the emotional story. Fleets are the business model — and we know the difference.

Photo: e-scooter battery-swap cabinets — Wikimedia Commons, CC BY-SA 4.0



What we have not proven yet

We would rather say it than be asked it.

A whole pack overheating uniformly

No deviation between cells, so Layer 1 stays silent. The hard 60 °C threshold covers it — not yet tested on hardware.

A real EV pack in service

So far: a lab pack of similar cells, plus public datasets. Measuring a vehicle in daily use is next.

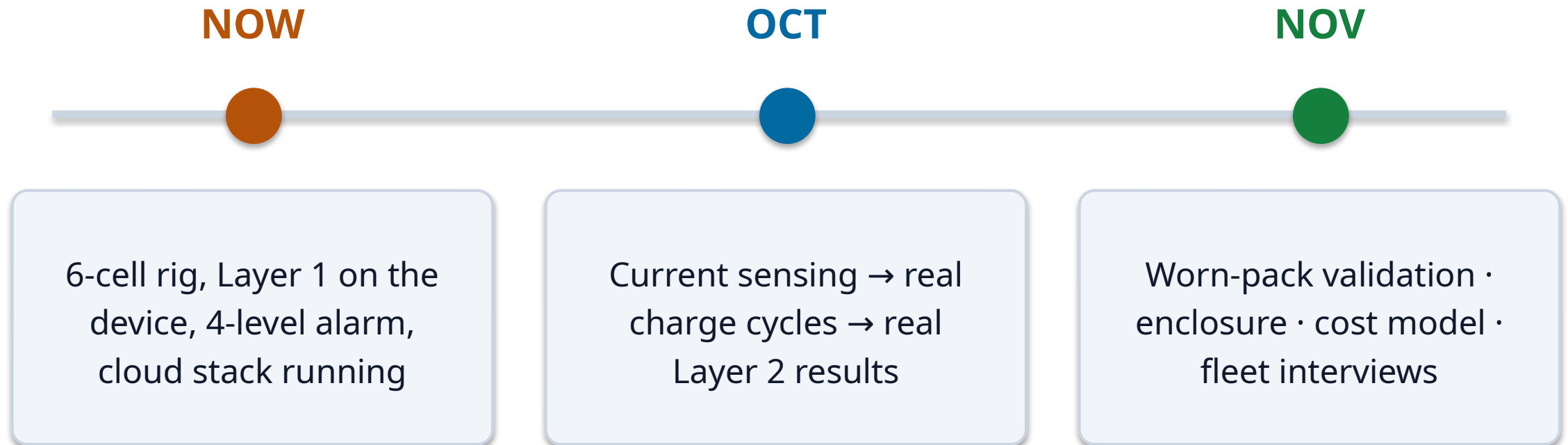
Ageing prediction on a worn pack

Our cells are new — one point at the top of the scale. We plan to borrow a degraded pack.

Nothing in this deck is claimed to be more than what we actually measured.

From semi-final to final

27 November 2026



One cell starts it. We watch every one of them.

**Team Hu Tieu · ICTU – Thai Nguyen
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Thank you. Questions welcome — especially the
hard ones.

Photo: 18650 cell after thermal runaway — Wikimedia Commons, CC BY 4.0

